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Attention Mechanism-Optimized Object Detection Algorithm for Traffic Scenes Based on YOLO26

Lijuan Shang
School of Artificial Intelligence
Neusoft Institute Guangdong
Foshan, China
shanglijuan@nuit.edu.cn

***Abstract*—With the rapid development of Intelligent Transportation Systems (ITS) and autonomous driving technologies, high-precision object detection in complex traffic scenarios has become crucial. To address the limitations of detection accuracy caused by drastic object scale variations, complex backgrounds, and severe occlusions in real-world traffic scenes, this paper proposes an attention mechanism-optimized object detection algorithm based on YOLO26, named YOLO26-LVCA. Building upon the state-of-the-art YOLO26 baseline, a novel Lightweight Voxel Coordinate Attention (LVCA) module is introduced. By employing direction-aware feature aggregation and low-dimensional bottleneck feature fusion, the LVCA module enables the network to adaptively focus on critical feature regions while effectively suppressing irrelevant background noise. This significantly enhances the feature representation capability for small and occluded objects. Extensive comparative experiments were conducted on the widely-used KITTI autonomous driving dataset. The results demonstrate that compared to the baseline YOLO26 model, the proposed algorithm achieves a mean Average Precision (mAP@0.5) of 87.2%, representing a 0.9% improvement, while maintaining a robust real-time detection speed of 126 FPS. Furthermore, Precision and Recall reached 91.7% and 82.6%, respectively. These experimental metrics fully validate the effectiveness and robustness of the proposed method in highly complex traffic environments.**

Keywords— Object detection, YOLO26, Attention mechanism, Traffic scenes, KITTI dataset, Autonomous driving

I. INTRODUCTION

TIn the research field of autonomous driving, environmental perception and decision planning are the cornerstones of intelligent advanced driver-assistance systems (ADAS). Accurate and real-time detection of traffic objects (such as vehicles, pedestrians, and cyclists) is the prerequisite for achieving reliable environmental perception. In recent years, deep learning-based object detection algorithms have made breakthrough progress, primarily branching into two-stage algorithms represented by Faster R-CNN and one-stage algorithms represented by the YOLO[1] (You Only Look Once) series. Due to their end-to-end network architecture and outstanding inference speed, YOLO algorithms have been widely adopted in both academia and industry. The recently proposed YOLO26 framework has further optimized the network architecture and feature fusion strategies, achieving an excellent balance between detection speed and accuracy.

However, in highly complex real-world traffic scenarios, existing detection models still face several severe challenges: Drastic Scale Variations: In a single field of view, the pixel dimensions of distant pedestrians and nearby buses differ enormously, demanding extreme multi-scale adaptability from the model. Severe Occlusions: Mutual occlusions among vehicles, or pedestrians hidden behind obstacles, frequently lead to the loss of critical features, thereby causing missed detections. Complex Backgrounds: Dynamic illumination changes, shadows, and cluttered urban architectural backgrounds easily interfere with foreground objects, resulting in false positives.

To address the aforementioned issues, this paper introduces a novel attention mechanism into the YOLO26 baseline. Attention mechanisms simulate the human visual system, guiding the network to allocate computational resources preferentially to more critical feature channels and spatial locations. The main contributions of this paper are summarized as follows: We design and propose a Lightweight Voxel Coordinate Attention (LVCA) module, which achieves spatial direction-aware feature perception with extremely low computational overhead.We construct an improved YOLO26-LVCA traffic object detection model by seamlessly integrating the LVCA module into the deep feature extraction stages of the backbone, effectively mitigating missed detections caused by occlusions and small targets.We conduct comprehensive experimental validation on the authoritative KITTI dataset[2], demonstrating the superiority of the improved algorithm in complex scenarios.

II. RELATED WORK

A. YOLO-based Object Detection

Since its inception, the YOLO series has undergone numerous iterations. From the early YOLOv1 to recent versions like YOLOv8, YOLOv10, and currently YOLO26[6], the core developmental trend has focused on designing more efficient Backbones, enhanced feature pyramid networks (Necks), and anchor-free Heads. YOLO26 further enhances comprehensive performance by introducing lightweight modules and novel loss functions. However, when faced with congested and highly occluded traffic conditions, its ability to focus on critical features and resist noise still has room for improvement.

B. Attention Mechanisms in Computer Vision

The application of attention mechanisms has greatly advanced various perception tasks. For instance, the Squeeze-and-Excitation (SE) module recalibrates channel features by modeling inter-channel dependencies[3]. The Convolutional

Block Attention Module (CBAM) combines channel and spatial attention for more comprehensive feature enhancement

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[4][5]. While integrating attention mechanisms into detection networks effectively boosts foreground perception without significantly increasing computational load, traditional mechanisms often neglect long-range spatial positional information. Therefore, this paper introduces a coordinate attention-based approach specifically optimized for traffic scenes.

III. Proposed Method

A. YOLO26 Architecture

The overall architecture of YOLO26 consists of four main components: the input end, the Backbone, the Neck, and the Head. The Backbone is responsible for extracting multi-scale feature maps from the input image. The Neck employs a bidirectional multi-scale feature fusion strategy (top-down and bottom-up). The Head outputs the class probabilities and bounding box coordinates of the objects. The YOLO26 network structure of this module is shown in Figure 1.

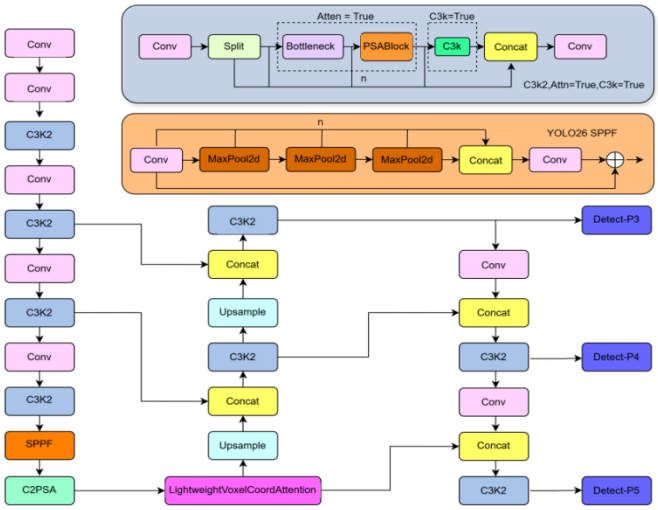


Fig. 1. Original network structure diagram of YOLO26

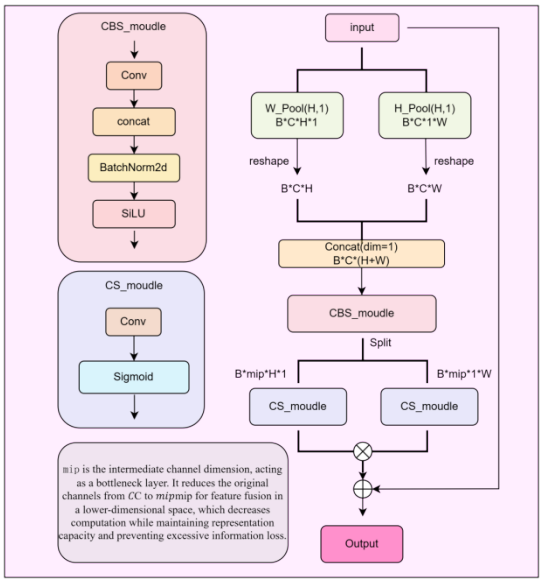


Fig. 2. Structure of the LVCA Module

B. Lightweight Voxel Coordinate Attention (LVCA)

To enhance the network's sensitivity to traffic targets while maintaining lightweight characteristics, this paper designs and introduces the LightweightVoxelCoordAttention (LVCA) module. The network structure of this module is shown in Figure 2.

The LVCA attention module designed in this paper can further process the feature maps processed by the YOLO26 backbone network. Its underlying concept and strategy are to capture the spatial feature dependencies in the feature map by performing one-dimensional feature aggregation on the input features along the horizontal and vertical directions respectively. The feature map first needs to pass through a directional pooling layer. During this period, two pooling kernels with sizes of （H，1）and（1，W）are used to pool along the height and width directions respectively, thereby extracting the one-dimensional feature information in the feature map. Specifically, the output of channel c at height h and width w can be expressed by Equation 1 and Equation 2 respectively:

$$z_c^h(h) = \frac{1}{W}\sum_{i=1}^{W} X_c(h,i), \tag{1}$$

$$z_c^w(w) = \frac{1}{H}\sum_{i=1}^{H} X_c(j,w) \tag{2}$$

In the formulas, c , h and w represent the number of channels, height, and width of the feature map, respectively. $z_c^w(w)$and $z_c^h(h)$represent the output feature values aggregated along the height and width directions, respectively, while X_c represents the c-th channel of the input feature map. After processing through these formulas, direction-aware feature maps in two directions are obtained. This operation is not only conducive to preserving the position information of height and width but also effectively captures long-distance contextual feature information.

Subsequently, after dimension adjustment, the features need to be concatenated and fused using a concat operation. The concatenated features are then fed into the CBS module, where the feature map undergoes convolution, batch normalization, and SiLU activation function processing. Its mathematical expression is shown in Equation 3:

$$F = \delta(B(Conv(Z))) \in R^{mip\times(H+W)} \tag{3}$$

In the formula, δ represents the SiLU activation function, and B represents Batch Normalization. $R^{mipx(H+W)}$ represents the tensor of the feature map. This module compresses the number of channels from c to the intermediate dimension mip(bottleneck layer). By performing feature fusion in a low-dimensional space, the computational complexity of the model is significantly reduced while preventing excessive loss of feature information.

Following the above operations, the fused feature information map F is split back into two parts. These two parts are then respectively input into independent CS modules to generate corresponding attention weights. This module consists of a 1x1 convolution and a Sigmoid activation function to restore the number of channels to c and output the weight distribution.

Finally, the generated attention weights in the horizontal and vertical directions are subjected to tensor multiplication to obtain

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the final two-dimensional spatial attention distribution, which is then applied to the original input feature X . To prevent the gradient vanishing problem caused by the increase in network depth, this paper introduces a residual connection to address this. This approach not only highlights critical feature regions but also effectively strengthens the convergence process of the network.

C. YOLO26-LVCA Network Architecture

This paper embeds the aforementioned LVCA attention module into the output end of the deep feature map of the YOLO26 backbone network. This layout strategy enables the network to pre-filter large-area background noise, such as trees and buildings, before multi-scale semantic interaction, and strengthens the feature response of key traffic participants, such as vehicles and pedestrians. Consequently, it effectively improves the detection accuracy on the KITTI dataset. The network structure is shown in Figure 3.

IV. Experiments And Analysis

A. Dataset Introduction

The experiments in this paper utilize the authoritative KITTI dataset in the field of autonomous driving. This dataset is collected from real urban roads, rural areas, and highway scenarios. We selected three major categories closely related to traffic in the dataset for evaluation: Car, Pedestrian, and Cyclist. The dataset was randomly divided into training, validation, and testing sets at a ratio of 8:1:1).

B. Experimental Environment and Parameter Settings

The experiments were conducted on a server equipped with an NVIDIA RTX 3090 graphics card (24GB VRAM), running the Ubuntu 20.04 operating system, and utilizing the PyTorch 2.0.1 deep learning framework. The parameter settings during model training were as follows: the input image size was uniformly resized to 640×640, the Batch Size was set to 16, the initial learning rate was 0.01, the momentum parameter was 0.937, and the SGD optimizer combined with a cosine annealing learning rate decay strategy was employed. The model was trained for a total of 300 Epochs..

C. Evaluation Metrics

This paper adopts common evaluation metrics in the field of object detection: Precision (P), Recall (R), and mean Average Precision (mAP). Among them, mAP@.50 represents the mean average precision when the Intersection over Union (IoU) threshold is 0.5, which is the core metric for measuring the overall detection performance of the model..

D. Results and Analysiss

To verify the effectiveness of the improved algorithm after adding the LVCA module, it was compared with the original YOLO26 model under identical experimental conditions. It was also compared with various classical object detection methods for traffic scenarios[8][9]. The specific results are shown in Table 1.

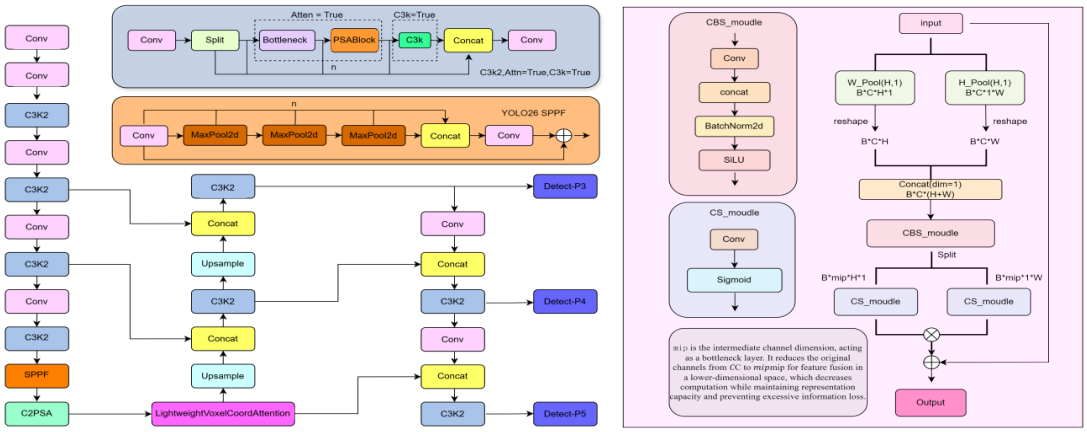


Fig. 3. YOLO26-LVCA Network Architecture

TABLE I. Performance Comparison of Different Detection Algorithms on the KITTI Datasetes

Method	Precision(%)	mAP@.50（%）	Recall(%)	FPS
Faster-RCNN	75.2	70.1	68.5	15
YOLOv5s	78.4	72.9	71.2	140
YOLOv8n	81.3	75.5	73.6	158
SRF-YOLOv7	83.4	77.4	75.4	107
RE-YOLOv5	84.9	78.9	77.8	93
YOLO26	88.2	85.8	78.4	145
YOLO26-CBAM	89.2	86.3	79.1	132
YOLO26- coordinate attention	90.1	86.7	80.1	130
YOLO26-LVCA without residual connection	91.3	86.9	81.7	128
YOLO26-LVCA	91.7	87.2	82.6	126

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As can be seen from the experimental results in Table 1, although the early two-stage model Faster R-CNN has acceptable accuracy, its inference speed is only 15 FPS, which cannot meet the requirements of autonomous driving. Mainstream YOLOv5s and YOLOv8n possess extremely high inference speeds but perform poorly in terms of recall, indicating that there are many missed detections in crowded or occluded scenarios. Recent cutting-edge algorithms such as RE-YOLOv5[7], which incorporates receptive field enhancement, achieved an mAP of 78.9%, but the introduction of complex mechanisms led to a significant drop in FPS to 93.

The YOLO26 + LVCA algorithm proposed in this paper demonstrates the best comprehensive performance. After introducing the LightweightVoxelCoordAttention mechanism, compared with the original YOLO26, the model achieved significant improvements in all core metrics: Precision increased by 3.5%, effectively reducing background false positives; Recall significantly increased by 4.2%, greatly improving the missed detection issue for occluded targets; and the overall mAP@.50 reached the highest level of 87.2%. Most importantly, benefiting from the unique low-dimensional bottleneck layer (mip) dimensionality reduction fusion design of the LVCA module, the algorithm achieved a massive leap in detection performance with only a slight decrease in inference speed (from 145 to 126 FPS), which still far exceeds the real-time requirements of autonomous driving systems.

The visualization results are shown in Figure 4, which from left to right are the detection results of Ground Truth, YOLO26, YOLO26-CBAM, and YOLO26-LVCA, respectively. This paper proves the effectiveness of the proposed method through the visualization results of YOLO26-LVCA and related models.



Fig. 4. Visual Comparison Results of Different Methods on the KITTI Dataset

V. Conclusion

Addressing the issues of insufficient detection accuracy caused by large target scale variations and severe occlusions in complex traffic scenarios, this paper proposes an improved YOLO26 algorithm based on LightweightVoxelCoordAttention (LVCA). By introducing this lightweight attention module into the network and utilizing direction-aware pooling and low-dimensional bottleneck feature fusion technology, the model significantly enhances its ability to extract key features of traffic targets while effectively suppressing background noise interference. Experiments on the KITTI dataset show that the improved algorithm substantially improves the detection accuracy of cars, pedestrians, and non-motorized vehicles with minimal computational overhead. Future work will attempt to apply this module to more complex 3D object detection tasks and further explore TensorRT deployment optimization on vehicle-mounted edge computing devices.

Acknowledgment

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References

[1] Redmon J, Divvala S, Girshick R, et al. You only look once: Unified, real-time object detection[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2016: 779-788.

[2] Geiger A, Lenz P, Urtasun R. Are we ready for autonomous driving? the kitti vision benchmark suite[C]//2012 IEEE conference on computer vision and pattern recognition. IEEE, 2012: 3354-3361..

[3] Hou Q, Zhou D, Feng J. Coordinate attention for efficient mobile network design[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2021: 13713-13722..

[4] Woo S, Park J, Lee J Y, et al. Cbam: Convolutional block attention module[C]//Proceedings of the European conference on computer vision (ECCV). 2018: 3-19..

[5] Hu J, Shen L, Sun G. Squeeze-and-excitation networks[C]//Proceedings of the IEEE conference on computer vision and pattern recognition. 2018: 7132-7141..

[6] Yang B, Hu Z. A Target Detection Algorithm for Auto Drive Based on SRF-YOLOv7[C]//2024 IEEE 7th Information Technology, Networking, Electronic and Automation Control Conference (ITNEC). IEEE, 2024, 7: 434-438.

[7] Li T, Xiong X, Zhang Y, et al. RE-YOLOv5: Enhancing Occluded Road Object Detection via Visual Receptive Field Improvements[J]. Sensors, 2025, 25(8): 2518..

[8] Yang M, Fan X. YOLOv8-Lite: A lightweight object detection model for real-time autonomous driving systems[J]. ICCK Transactions on Emerging Topics in Artificial Intelligence, 2024, 1(1): 1-16.

[9] Tang L, Yun L, Chen Z, et al. HRYNet: a highly robust YOLO network for complex road traffic object detection[J]. Sensors, 2024, 24(2): 642.

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